

Unit 13, Lesson 4: Comparing Graphical Representations

Benchmark

MA.7.DP.1.2 Given two numerical or graphical representations of data, use the measure(s) of center and measure(s) of variability to make comparisons, interpret results, and draw conclusions about the two populations.

Learning Target

- ☐ I can use two graphical representations of data to make comparisons and draw conclusions about the populations.

13.4.1 Warm-Up: Defrost Time

1. A microwave shows that it takes 9 minutes and 45 seconds to defrost 1.5 pounds of meat and 26 minutes to defrost 4 pounds of meat.
 - a. Without performing any calculations, estimate a range of minutes you think it will take to defrost 3.5 pounds of meat.
 - b. Explain how the given information could be used to find the exact amount of time it takes to defrost 3.5 pounds of meat.
 - c. Determine how long it would take for the microwave to defrost 3.5 pounds of meat. Show your work.

13.4.2 Guided Practice: Comparing Line Plots

1. Clare recorded the amount of time a sample of students in the sixth, eighth, and tenth grades spent doing homework, in hours per week. She made a line plot of the data for each grade and provided the following summary:
- Students in sixth grade tend to spend less time on homework than students in eighth and tenth grades.
 - The homework times for the tenth-grade students are more alike than the homework times for the eighth-grade students.

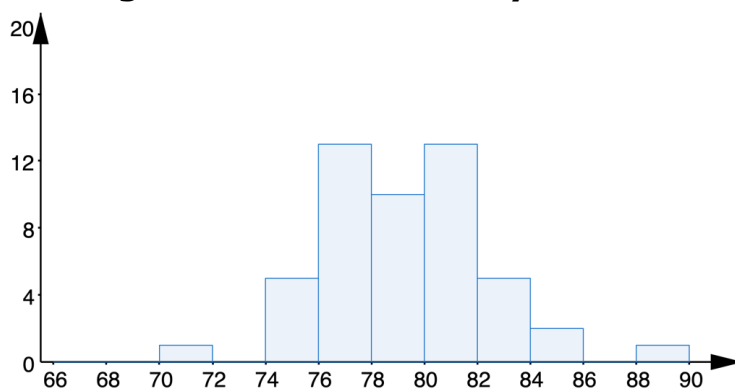
Use Clare's summary to match each line plot to the correct grade (sixth, eighth, or tenth).

Grade Level	Line Plot

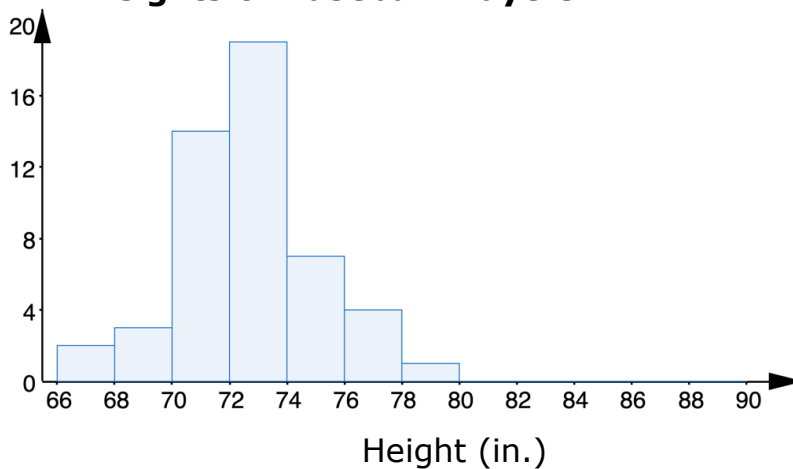
13.4.3 Activity: Station 1 – Histograms and Line Plots

1. The two histograms show height distributions of 50 male professional baseball players and 50 male professional basketball players, in inches (in.).

Heights of Basketball Players



Heights of Baseball Players



- a. Complete the statements that draw a conclusion about the typical height of professional basketball players and baseball players.

The sample data in the histograms shows the

center of the data for basketball players is

☐ less than

☐ greater than

the center of the data for baseball

players. It can be concluded that professional

basketball players are typically

☐ shorter than

☐ taller than

professional baseball players.

- b. Complete the statement.

The histograms show that in the samples, the

range of heights of basketball players is

☐ less than

☐ greater than

the range of the heights of

baseball players. We can conclude that professional

basketball players have

☐ more

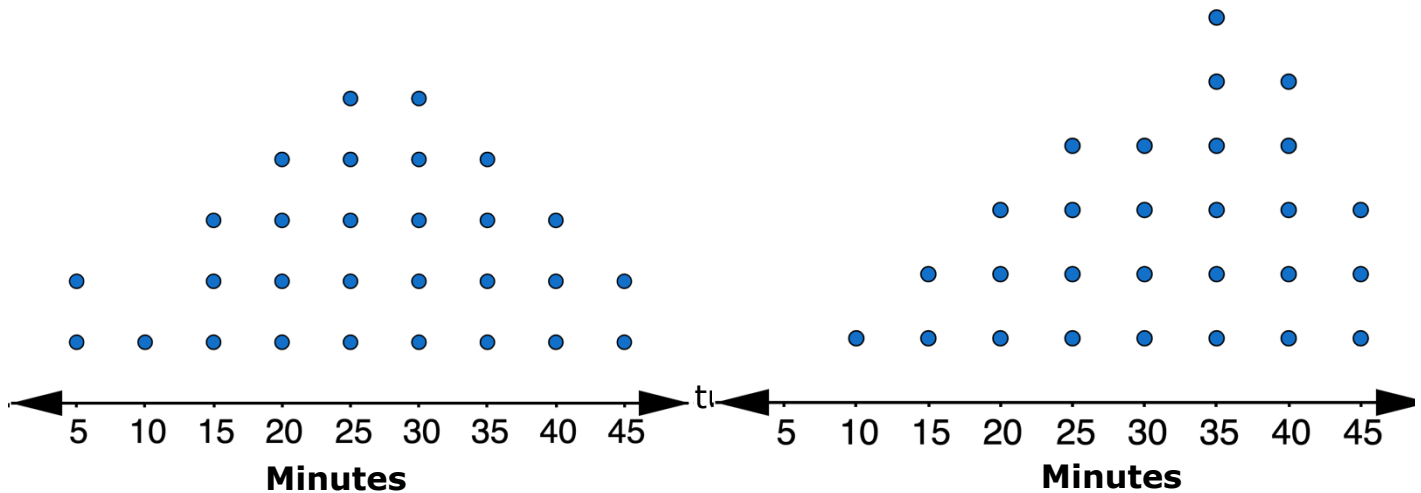
☐ less

variability in

their heights than professional baseball players.

One possible reason for this is ...

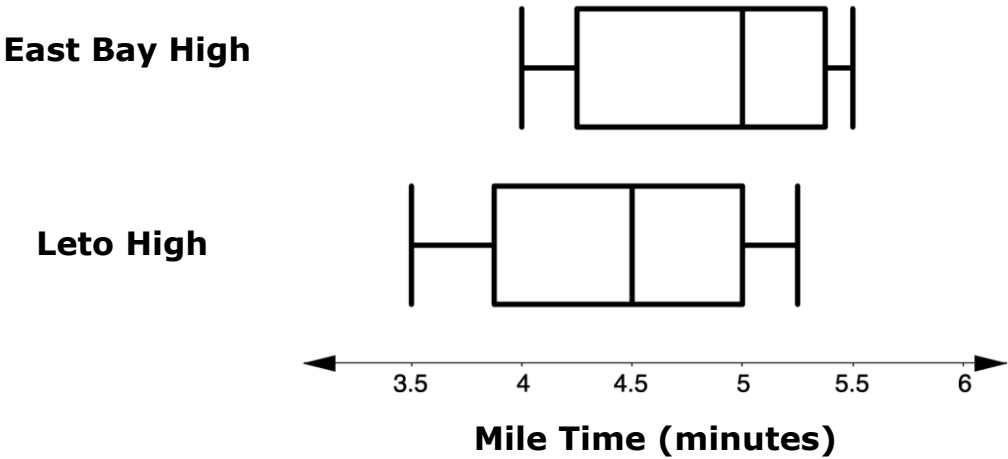
2. Mr. Logan surveyed a sample of his junior and senior students about the time they spend studying math per day. He then tabulated the results and created a line plot displaying the data for each group.

Time Spent Studying (Juniors)**Time Spent Studying (Seniors)**

- b. The value of the larger mean study time for the two groups is _____.
- c. In one to two sentences, describe the difference between the number of minutes Mr. Logan's juniors and seniors study by comparing the center and shapes for the groups.

13.4.4 Activity: Station 2 – Box Plots

1. The mile times for a sample of track team members at East Bay High School and Leto High School are shown in the two box plots.



- a. Using the box plots above, calculate and describe each of the features for Leto High in the table.

Feature	East Bay High School	Leto High School
Shape	The shape of the data is asymmetrical.	
Center	The median mile time for the sample of East Bay High team members is 5 minutes.	
Variability	The range of the sample of East Bay High team members' mile times is 1.5 minutes. The IQR of the sample of East Bay High team members' mile times is 0.75 minutes.	

- b. Based on the data, explain which team is more likely to win at a track meet.
- c. Suppose a runner at a track meet runs a mile in 3 minutes and 45 seconds. Explain which track team you would assume they were on, using information from the data displays.
- d. Based on your conclusions and answers to Parts A through C, use some of the terms from the word bank below to write a 4-5 sentence summary comparing the population of runners at East Bay High School to the population of runners at Leto High School.

mean

median

range

IQR

center

variability

upper quartile

lower quartile

sample

population

13.4.5 Activity: Station 3 – Stem-and-Leaf Plots and Box Plots

1. A sample of two stores' guitar prices are shown in the stem-and-leaf plots.

Store A's Guitar Prices

Stem	Leaf
2	1, 2, 8
3	3, 6
4	5, 8, 8
5	1, 6
6	
7	2, 2
8	1, 6
9	1

Key: 2|1 = \$210.00

Store B's Guitar Prices

Stem	Leaf
4	5, 8
5	6, 9
6	1, 1, 3, 5
7	5, 5, 6
8	0, 6
9	1, 8
10	0

Key: 2|1 = \$210.00

Zach wants to purchase a guitar and is trying to determine which store to purchase from. Using the displays he makes the following conclusion.

"Store B is better for more experienced guitar players because it typically sells more expensive guitars than Store A. The median price of Store B's guitars is \$700, while the median price of Store A's guitars is \$480. Prices of guitars at Store A tend to vary more, with an IQR of \$390, compared to the prices of guitars at Store B, with a much smaller IQR."

Do you agree or disagree with Zach's conclusion? Explain your reasoning.

2. Suppose a sample of test scores is taken from Mr. Spellman’s morning math classes. The data is displayed in the top box plot. Another sample of test scores is taken from Mr. Spellman’s afternoon math classes. The data is displayed in the bottom box plot.



Explain the difference between taking Mr. Spellman’s math test in the morning compared to the afternoon. Use some of the terms from the word bank and be sure to include both a measure of center and variability in your explanation.

mean	median	range	IQR
center	variability	upper quartile	
lower quartile	sample	population	

13.4.6 Wrap-Up: Reflection

Complete each statement based on your current understanding.

1. I am still unsure about _____

2. One question I have that will help me better understand is

?

